

GENERAL INFORMATION

DIAGNOSTIC TROUBLE CODE INDEX - DTC: REAR
DIFFERENTIAL CONTROL MODULE [RDCM] [G1557418]

DESCRIPTION AND OPERATION

REAR DIFFERENTIAL CONTROL MODULE [RDCM]

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle

NOTES:

- Guided Diagnostics must be followed and the repair advised by the process completed. Failure to do so may result in the rejection of any warranty claim made.
- If the control module or a component is at fault and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval program is in operation, prior to the installation of a new module/component
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system)
- When performing voltage or resistance tests, always use a digital multimeter that has the resolution ability to view 3 decimal places. For example, on the 2 volts range can measure 1mV or 2 K Ohm range can measure 1 Ohm. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals
- If the rear differential control module, rear differential locking motor or the rear differential are replaced, the on demand self test (ODST) must be carried out to calibrate the components
- Check JLR claims submission system for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and complete the recommendations as required.

The table below lists all diagnostic trouble codes (DTCs) that could be set in the rear differential control module, for additional diagnosis and testing information refer to the relevant diagnosis and testing section. For additional information, refer to: [Rear Drive Axle and Differential](#) (205-02 Rear Drive Axle/Differential, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0562-16	System Voltage Low - Circuit voltage below threshold	■ Rear differential control module - Supply voltage below 9 volts ■ Rear differential control module failure	■ Clear DTC and retest ■ If fault persists, complete the guided diagnostic routine using the Jaguar Land Rover Approved Diagnostic Equipment
P0563-17	System Voltage High - Circuit voltage above threshold	■ Rear differential control module - Supply voltage above 16 volts	■ Refer to the electrical circuit diagrams and check the power and ground circuits to the module. Repair circuit as required. Clear DTC and retest ■ Check engine control module for stored voltage-related DTCs to confirm if the vehicle has a power supply issue and refer to the relevant DTC index ■ Refer to relevant section of workshop manual and battery care manual. Check battery state of charge and starting/charging system performance
P0604-00	Internal Control Module Random Access Memory (RAM) Error - No sub type information	■ Rear differential control module - Software fault ■ Rear differential control module failure	■ Clear DTC and retest. If fault persists, check and install latest relevant level of software to the rear differential control module using the Jaguar Land Rover Approved Diagnostic Equipment. Clear DTC and retest ■ If fault persists, check and install a new rear differential control module as required
P0605-00	Internal Control Module Read Only Memory (ROM) Error - No sub type information	■ Rear differential control module - Software fault ■ Rear differential control module failure	■ Clear DTC and retest. If fault persists, check and install latest relevant level of software to the rear differential control module using the Jaguar Land Rover Approved Diagnostic Equipment. Clear DTC and retest ■ If fault persists, check and install a new rear differential control module as required
P0606-47	Control Module Processor - Watchdog / safety MicroController failure	■ Watchdog reset - Rear differential control module failure	■ This is a control module internal check DTC and is not necessarily a fault. If no other DTCs are set and no customer complaint exists, clear DTC and retest ■ Check the security of control module fixings. Check the module ground connection. Consider environmental conditions before condemning the rear differential control module ■ Clear DTC and retest. If fault persists, check and install latest relevant level of software to the rear differential control module using the Jaguar Land Rover Approved Diagnostic Equipment. Clear DTC and retest ■ If the problem persists, check and install a new rear differential control module as required
P0607-49	Control Module Performance - Internal electronic failure	■ Rear differential control module failure - Charge pump voltage below threshold	■ Check engine control module for stored DTCs and refer to the relevant DTC index ■ Refer to relevant section of workshop manual and battery care manual. Check battery state of charge and starting/charging system performance ■ Refer to the electrical circuit diagrams and check the power and ground circuits to the module. Repair circuit as required. Clear DTC and retest ■ If fault persists, check and install a new rear differential control module as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P062F-00	Internal Control Module EEPROM Error - No sub type information	Memory error - Rear differential control module failure	- Cycle ignition, clear DTC and retest. If fault persists, check and install latest relevant level of software to the rear differential control module using the Jaguar Land Rover Approved Diagnostic Equipment - If module software is updated and the fault persists, check and install a new rear differential control module as required
P0652-16	Sensor Reference Voltage B Circuit Low - Circuit voltage below threshold	- Rear differential locking motor position sensor supply below 5.7 volts - Sensor failure (within rear differential locking motor)	- Refer to the electrical circuit diagrams and check the rear differential locking motor position sensor circuit, repair as necessary. Clear the DTC and retest - If no circuit problems exist, check and install a new rear differential locking motor as required
P0653-17	Sensor Reference Voltage B Circuit High - Circuit voltage above threshold	- Rear differential locking motor position sensor supply above 8.3 volts - Rear differential control module failure	- This is a control module internal check DTC. If no other DTCs are set and no customer complaint exists, clear this DTC and retest - Check the security of control module fixings. Check the module ground connection. Consider environmental conditions before condemning the control module - Refer to the electrical circuit diagrams and check the rear differential locking motor hall sensor reference voltage at the control module and at the rear differential locking motor - If voltage at the rear differential locking motor is too high, check for short circuit fault on the rear differential locking motor hall sensor supply circuit. Repair circuit as required. Clear DTC and retest - If voltage at the module is too high, then suspect a control module fault - Clear DTC and retest. If fault persists, check and install a new rear differential control module as required
P0657-13	Actuator Supply Voltage A Circuit / Open - Open circuit	NOTE: If this DTC has set together with a ROM error DTC (P0605-00), the ROM error is the likely root cause. Refer to the DTC helptext for DTC P0605-00 and rectify this fault first before proceeding further - Rear differential control module - Software fault - Rear differential control module failure - Rear differential locking motor supply circuit - Open circuit, high resistance	NOTE: If this DTC has set together with a ROM error DTC (P0605-00), the ROM error is the likely root cause. Refer to the DTC helptext for DTC P0605-00 and rectify this fault first before proceeding further Refer to the electrical circuit diagrams and check the rear differential locking motor clutch actuation motor supply circuit for open circuit, high resistance. Repair circuit as required. Clear DTC and retest
P0658-11	Actuator Supply Voltage A Circuit / Low - Circuit short to ground	Rear differential locking motor supply circuit - Short circuit to ground	Refer to the electrical circuit diagrams and check the rear differential locking motor clutch actuation motor supply circuit for short circuit to ground. Repair circuit as required. Clear DTC and retest
P0659-12	Actuator Supply Voltage A Circuit / High - Circuit short to battery	Rear differential locking motor clutch actuation motor supply circuit - Short circuit to power	Refer to the electrical circuit diagrams and check the rear differential locking motor clutch actuation motor supply circuit for short circuit to power. Repair circuit as required. Clear DTC and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P080A-54	Clutch Position Not Learned - Missing calibration	■ Rear differential control module not calibrated ■ Rear differential control module failure ■ Differential clutch stuck or out of tolerance	NOTE: The rear differential control module system runs a self calibration routine every 500Km, if any other rear differential control module fault is present this routine will not run and log P080A. If other rear differential control module related faults are also set these should be investigated first. P080A will also log if the rear differential control module has been removed and reconnected ■ Clear DTCs and complete the on-demand self test through the 'Monitor DTC' tab on the manufacturer approved diagnostic system ■ Clear DTCs, ignition state to off for 5 seconds then retest ■ If fault persists, check and install a new rear differential control module as required
P0810-62	Clutch Position Control Error - Signal compare failure	■ Mismatch of actual and expected/calculated rear differential locking motor position ■ Signal circuit fault ■ Sensor fault ■ Rear differential control module failure	NOTE: The position sensor is internal to the rear differential locking motor ■ Record then clear any stored DTCs ■ Complete the on-demand self test through the 'Monitor DTC' tab on the Jaguar Land Rover Approved Diagnostic Equipment ■ Clear DTCs, ignition state to off for 5 seconds, roadtest vehicle, then retest ■ If fault persists, complete the guided diagnostic routine using the Jaguar Land Rover Approved Diagnostic Equipment ■ Clear DTCs, ignition state to off for 5 seconds then retest ■ If fault persists, check and install a new rear differential locking motor as required
POCBC-11	Drive Motor "A" Temperature Sensor Circuit - Circuit short to ground	■ Rear differential locking motor internal temperature sensor circuit - Short circuit to ground, open circuit, high resistance ■ Rear differential locking motor failure	■ Refer to the electrical circuit diagrams and check rear differential locking motor temperature sensor circuit for short circuit to ground, open circuit, high resistance. Repair circuit as required. Clear DTC and retest ■ If no circuit problems exist or if fault persists, check and install a new rear differential locking motor as required
POCBC-15	Drive Motor "A" Temperature Sensor Circuit - Circuit short to battery or open	■ Rear differential locking motor internal temperature sensor circuit - Short circuit to power, open circuit, high resistance ■ Rear differential locking motor failure	■ Refer to the electrical circuit diagrams and check rear differential locking motor temperature sensor circuit for short circuit to power, open circuit, high resistance. Repair circuit as required. Clear DTC and retest ■ If no circuit problems exist or if fault persists, check and install a new rear differential locking motor as required
P186A-13	Differential Lock-up Actuator Brake Control Circuit / Open - Circuit open	■ Rear differential locking motor actuator brake solenoid circuit - Open circuit, high resistance ■ Rear differential locking motor failure	■ Refer to the electrical circuit diagrams and check rear differential locking motor actuator brake solenoid circuit for open circuit, high resistance. Repair circuit as required. Clear DTC and retest ■ If no circuit problems exist or if fault persists, check and install a new rear differential locking motor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P186B-11	Differential Lock-up Actuator Brake Control Circuit Low - Circuit short to ground	■ Rear differential locking motor actuator brake solenoid circuit - Short circuit to ground ■ Rear differential locking motor failure	■ Refer to the electrical circuit diagrams and check rear differential locking motor actuator brake solenoid circuit for short circuit to ground. Repair circuit as required. Clear DTC and retest ■ If no circuit problems exist or if fault persists, check and install a new rear differential locking motor as required
P186C-12	Differential Lock-up Actuator Brake Control Circuit High - Circuit short to battery	■ Rear differential locking motor actuator brake solenoid circuit - Short circuit to power ■ Rear differential locking motor failure	■ Refer to the electrical circuit diagrams and check rear differential locking motor actuator brake solenoid circuit for short circuit to power. Repair circuit as required. Clear DTC and retest ■ If no circuit problems exist or if fault persists, check and install a new rear differential locking motor as required
P186D-97	Clutch Actuator Stuck - Component or system operation obstructed or blocked	■ Rear differential locking motor clutch actuation motor circuit fault ■ Rear differential locking motor actuator brake fault ■ Actuating mechanism mechanical failure ■ Rear differential locking motor failure	NOTE: If any rear differential locking motor actuator brake or motor position sensor faults are also set these should be investigated first ■ Refer to the electrical circuit diagrams and check the resistance of the rear differential locking motor clutch actuation motor circuit (resistance should be within the range of 1.8 Ohms). Repair circuit as required. Clear DTC and retest ■ If resistance is within range, clear DTCs then complete the guided diagnostic routine using the manufacturers approved diagnostic system. Clear DTCs, cycle ignition and retest ■ If fault persists, check and install a new rear differential locking motor as required
P1888-11	Differential Oil Temperature Sensor Circuit Failure - Circuit short to ground	■ Rear differential oil temperature sensor circuit - Short circuit to ground	■ Refer to the electrical circuit diagrams and check the rear differential oil temperature sensor circuit for short circuit to ground. Repair circuit as required. Clear DTC and retest
P1888-15	Differential Oil Temperature Sensor Circuit Failure - Circuit short to battery or open	■ Rear differential oil temperature sensor circuit - Short circuit to power, short circuit to one another, open circuit, high resistance	■ Refer to the electrical circuit diagrams and check the rear differential oil temperature sensor circuit for short circuit to power, short circuit to one another, open circuit, high resistance. Repair circuit as required. Clear DTC and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P188A-98	Differential Oil Temperature Too High/Too Low - Component or system over temperature	NOTE: If this DTC has set together with a ROM error DTC (P0605-00), the ROM error is the likely root cause. Refer to the DTC helptext for DTC P0605-00 and rectify this fault first before proceeding further ■ Rear differential oil temperature sensor value above 160°C ■ Insufficient oil quantity ■ Incorrect oil specification ■ Rear differential oil temperature sensor failure	NOTE: If this DTC has set together with a ROM error DTC (P0605-00), the ROM error is the likely root cause. Refer to the DTC helptext for DTC P0605-00 and rectify this fault first before proceeding further ■ Check the operation of the rear differential. Refer to the relevant section of the workshop manual and check the rear differential oil quantity and specification. Rectify as required, clear DTC and retest ■ If fault persists, clear DTC, allow the vehicle to cool for at least 1 hour, then retest. If DTC persists, refer to the electrical circuit diagrams and check the rear differential oil temperature sensor connectors and circuits for short circuit to ground, short circuit to power, short circuit to one another, open circuit, high resistance. Repair circuit as required. Clear DTC and retest ■ If no circuit problems exist or if fault persists, check and install a new rear differential oil temperature sensor as required
P18A5-12	Motion Position Sensor "A" Signal Circuit - Circuit short to battery	■ Rear differential locking motor position sensor signal circuit - Short circuit to power ■ Sensor failure (within rear differential locking motor)	■ Refer to the electrical circuit diagrams and check the rear differential locking motor position sensor signal circuit for short circuit to power. Repair circuit as required. Clear DTC and retest ■ If no circuit problems exist or if fault persists, check and install a new rear differential locking motor as required
P18A5-14	Motion Position Sensor "A" Signal Circuit - Circuit short to ground or open	■ Rear differential locking motor position sensor circuit - Short circuit to ground, open circuit, high resistance ■ Sensor failure (within rear differential locking motor)	■ Refer to the electrical circuit diagrams and check the rear differential locking motor position sensor circuit for short circuit to power. Repair circuit as required. Clear DTC and retest ■ If no circuit problems exist or if fault persists, check and install a new rear differential locking motor as required
P18A8-74	Differential Actuator Brake Control - Actuator slipping	■ Internal actuator brake of rear differential locking motor is slipping ■ Rear differential locking motor failure	■ If resistance is within range (resistance should be within the range of 1.8 Ohms), clear DTCs then complete the guided diagnostic routine using the manufacturers approved diagnostic system ■ Clear DTC and retest. If fault persists, check and install a new rear differential locking motor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P2787-4B	Clutch Temperature Too High - Over temperature	NOTES: ■ If this DTC has set together with a ROM error DTC (P0605-00), the ROM error is the likely root cause. Refer to the DTC helptext for DTC P0605-00 and rectify this fault first before proceeding further ■ If this DTC is set alongside other temperature related system faults, then these should be rectified first ■ Rear differential locking motor internal temperature sensor (calculated from motor and oil temperature sensor values) value above 200°C ■ Extensive prolonged rear wheel speed differential ■ Rear differential oil temperature sensor circuit fault ■ Rear differential locking motor temperature sensor circuit fault ■ Insufficient oil quantity ■ Incorrect oil specification ■ Rear differential fault ■ Rear differential control module failure	NOTES: ■ If this DTC has set together with a ROM error DTC (P0605-00), the ROM error is the likely root cause. Refer to the DTC helptext for DTC P0605-00 and rectify this fault first before proceeding further ■ If this DTC is set alongside other temperature related system faults, then these should be rectified first ■ Check for related DTCs and investigate these first with reference to the relevant DTC index. Clear DTCs and retest ■ Confirm the customer concern and vehicle usage at the time of DTC event as this DTC could be triggered by excessive wheel slip due to prolonged off road use ■ Check the operation of the rear differential. Refer to the relevant section of the workshop manual and check the rear differential oil quantity and specification. Rectify as required, clear DTC and retest ■ Clear DTC and retest. If fault persists, check and install a new rear differential locking motor as required ■ Clear DTC and retest. If fault persists, check and install a new rear differential control module as required
U0001-81	High Speed CAN Communication Bus - Invalid serial data received	■ Unexpected data has been received through the high speed CAN network ■ High speed CAN fault	■ Refer to the electrical circuit diagrams and check the power and ground connections to the module ■ Using the Jaguar Land Rover Approved Diagnostic Equipment, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check the CAN network ■ Using the Jaguar Land Rover Approved Diagnostic Equipment, identify the module(s) responsible for the erroneous CAN signals. Check these modules for related DTCs and refer to the relevant DTC index
U0001-87	High Speed CAN Communication Bus - Missing message	■ Unexpected data has been received through the high speed CAN network ■ High speed CAN fault	■ Refer to the electrical circuit diagrams and check the power and ground connections to the module ■ Using the Jaguar Land Rover Approved Diagnostic Equipment, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check the CAN network ■ Using the Jaguar Land Rover Approved Diagnostic Equipment, identify the module(s) responsible for the erroneous CAN signals. Check these modules for related DTCs and refer to the relevant DTC index
U0001-88	High Speed CAN Communication - Bus off	■ High speed CAN bus off detected	■ Refer to the electrical circuit diagrams and check the power and ground connections to the module. Using the Jaguar Land Rover Approved Diagnostic Equipment, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check the CAN network

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U0300-00	Internal Control Module Software Incompatibility - No sub type information	Rear differential control module not configured - Incorrect software installed	Check and confirm correct rear differential control module is installed. If correct, clear DTC, ignition off for 5 seconds, ignition on and then check if DTC reoccurs. If the problem persists, configure the module using the Jaguar Land Rover Approved Diagnostic Equipment
U1A14-49	CAN Initialization Failure - Internal electronic failure	- Rear differential control module has failed the CAN serial data communication software initialization - Internal control module failure	- Clear the DTC and retest. If the problem persists, configure the module using the Jaguar Land Rover Approved Diagnostic Equipment. Clear the DTC and retest - If the problem persists, check and install a new rear differential control module as required
U2000-4B	Motor Temperature - Over temperature	NOTES: - Circuit reference - MAG+ - - Circuit reference - MAG- - - Rear differential locking motor temperature too high - Rear differential control module is not configured correctly - Differential locking motor internal failure	NOTES: - This DTC is set when the temperature of the differential locking motor exceeds 150°C. - This DTC may be induced by excessive electric differential clutch operation due to prolonged off road or track driving. - If other differential locking motor brake DTCs are set, perform the corrective actions for those DTCs first and retest. - After installing a new rear differential control module, differential locking motor or electric differential, clear the DTCs and calibrate the system by performing routine - On Demand Self Test. - Allow the system temperature to return to normal. Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest - Using the Jaguar Land Rover approved diagnostic equipment, reconfigure the rear differential control module with the latest level software and perform routine - On Demand Self Test. - If the fault persists, install a new differential locking motor
U2011-1C	Motor - Circuit voltage out of range	- Rear differential locking motor clutch actuation motor circuit - Short circuit to power, short circuit to each other, open circuit, high resistance - Rear differential locking motor failure	- Refer to the electrical circuit diagrams and check the rear differential locking motor clutch actuation motor circuit for short circuit to power, short circuit to each other, open circuit, high resistance. Repair circuit as required. Clear DTC and retest - If fault persists, check and install a new rear differential locking motor as required
U210A-62	Temperature Sensor - Signal compare failure	- Mismatch between values received from rear differential locking motor internal temperature sensor and the rear differential oil temperature sensor - Insufficient oil quantity - Rear differential locking motor internal temperature sensor circuit - Short circuit to ground, short circuit to power, open circuit, high resistance - Sensor fault - Rear differential locking motor failure	- Refer to the relevant section of the workshop manual and check the rear differential oil quantity and specification. Rectify as required, clear DTC and retest - Check the integrity of the connectors and wiring on both the rear differential locking motor internal temperature sensor circuit and the rear differential oil temperature sensor circuit. Repair circuit(s) as required. Clear DTC and retest - If fault persists, allow vehicle to cool for at least 1 hour, then read and compare temperature values from rear differential locking motor internal temperature sensor and the rear differential oil temperature sensor using the Jaguar Land Rover Approved Diagnostic Equipment (the temperature difference should be less than 25°C). If the reported temperature difference is greater than 25°C, check and install a new rear differential oil temperature sensor and/or a new rear differential locking motor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U3000-4B	Control Module - Over temperature	■ Rear differential control module - Overheat detected (internal temperature sensor value greater than 105°C) ■ Rear differential control module - Insecurely mounted ■ Rear differential control module failure	■ Check the security and integrity of the rear differential control module fixings. Rectify or repair as required. Clear DTC and retest ■ If fault persists, allow vehicle to cool for at least 1 hour, then recheck temperature readings. If module continues to report an over temperature state after cooling, check and install a new rear differential control module as required

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